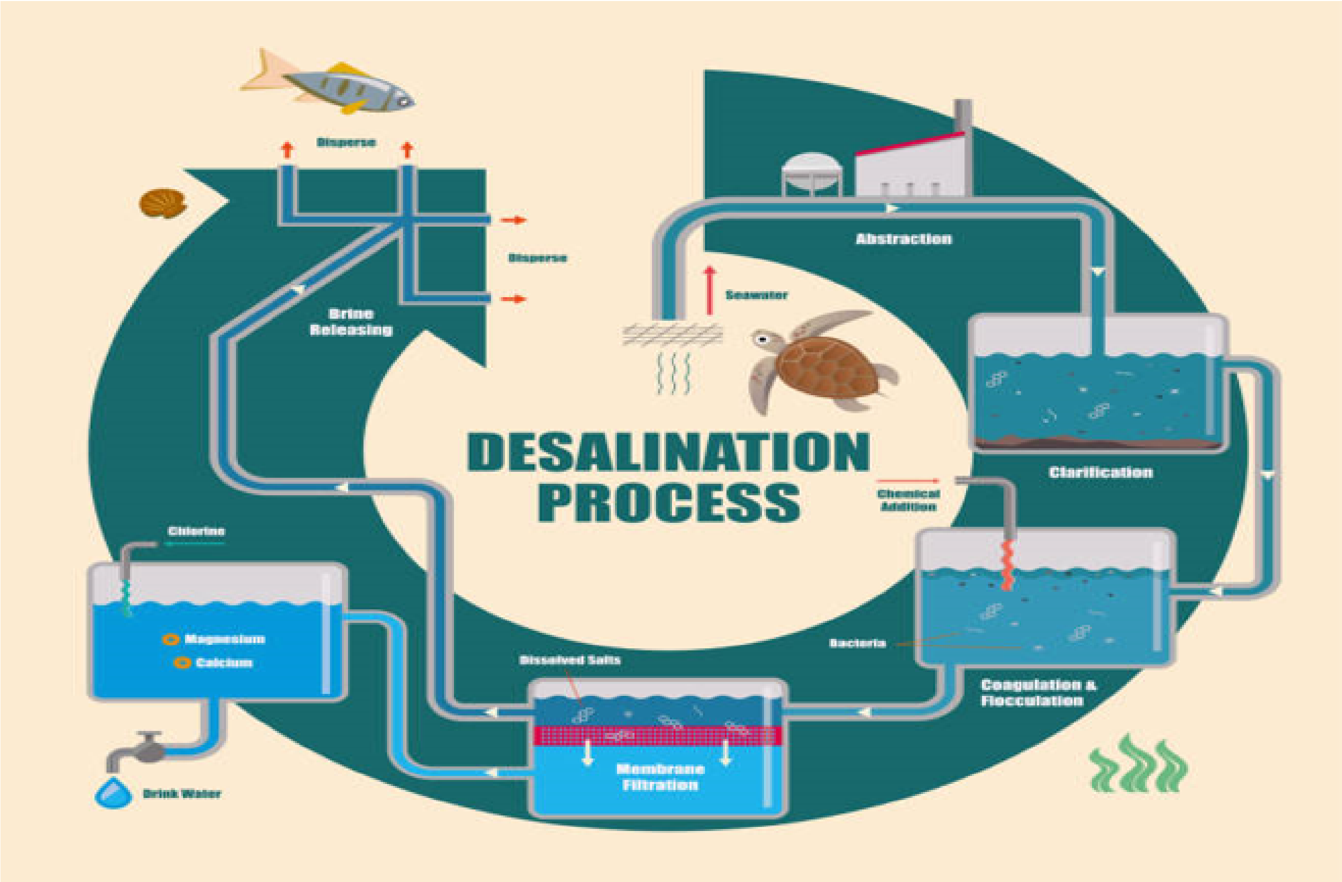
Qualification Pack



Green Hydrogen Plant Junior Technician (Desalination)

QP Code: SGJ/Q4303

Version: 1.0

NSQF Level: 3

Skill Council for Green Jobs || 3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri
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Contents

SGJ/Q4303: Green Hydrogen Plant Junior Technician (Desalination)

Brief Job Description

Green Hydrogen Plant Junior Technician-Desalination would install Mechanical and Electrical equipment of Desalination Unit in green hydrogen plant unit along with associated civil works. He/She would assist in performing Pre-commissioning checks/tests and subsequent commission, operation and maintenance of desalination unit in green hydrogen/green ammonia plant.

Personal Attributes

This job requires the individual to concentrate on the job at hand and complete the work safely. He/she should be able to communicate in local language. He / She must possess energy and strength for physical work. And also have a basic understanding of social and natural environment. He/she should possess very good interpersonal skills to work in a desalination unit of a green hydrogen/green ammonia plant.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. SGJ/N4301: Basics of Green Hydrogen Production
2. SGJ/N4302: Analyse Main Parts of green hydrogen production unit
3. SGJ/N4311: Basics of Desalination process and Its Importance for Green Hydrogen Production
4. SGJ/N4312: Perform pretreatment of water before Desalination for green hydrogen production
5. SGJ/N4313: Quality of water feed to the electrolyser
6. SGJ/N4314: Perform Operation & Maintenance of desalination unit
7. SGJ/N4315: Maintain Health & Safety at Desalination unit of Green Hydrogen generation project site
8. DGT/VSQ/N0101: Employability Skills (30 Hours)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician

Country	India
NSQF Level	3
Credits	12
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 8131.2100
Minimum Educational Qualification & Experience	10th grade pass with NA of experience OR 8th Class (withwith 2-years of (NTC /NAC) after 8th) with NA of experience OR Previous relevant Qualification of NSQF Level (NSQF Level 2.5 with) with 1-2 Years of experience
Minimum Level of Education for Training in School	8th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/08/2026
NSQC Approval Date	31/08/2023
Version	1.0
Reference code on NQR	QG-03-ES-00766-2023-V1-SCGJ
NQR Version	1

Remarks:

Total 360 Hours i.e. 12 Credits (Theory: 170 hours+Practical:100 hours+ 30 hours of employability skills + 60 hours of OJT)

SGJ/N4301: Basics of Green Hydrogen Production

Description

This unit specified the fundamentals of green hydrogen production system

Scope

The scope covers the following :

- Basics of Green Hydrogen Production

Elements and Performance Criteria

Basics of Green Hydrogen Production

To be competent, the user/individual on the job must be able to:

PC1. discuss properties and characteristics of Hydrogen

PC2. describe basic concepts of Hydrogen as fuel and energy carrier

PC3. discuss in brief various existing methods of hydrogen production and demonstrate various methods of hydrogen production.

PC4. discuss various colour code nomenclature of Hydrogen and demonstrate with chart colour code nomenclature of Hydrogen

PC5. discuss various technology options for production of Green Hydrogen and draw a flow diagram of green hydrogen production, conversion and end uses across the energy system

PC6. discuss key aspects and challenges related to production and storage of Green Hydrogen and do an activity for matching the process and source of production as applicable for different colour codes of hydrogen

PC7. discuss the end use applications of Green hydrogen in industry, transport and power production

PC8. discuss the role and responsibilities of various Green Hydrogen Plant Junior Technicians and demonstrate roles of various technicians in a green hydrogen plant

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organizations reporting structure

KU2. organizations documentation policy

KU3. organizational culture

KU4. Properties and characteristic of hydrogen

KU5. concept of hydrogen as fuel and energy carrier

KU6. signs, symbols and color codes and nomenclature used in hydrogen production

KU7. challenges related to production and storage of green hydrogen

KU8. basic skills required to perform the task of green hydrogen production

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. fill up relevant documents.
GS2. read vernacular language.
GS3. understand the various colour codes used in hydrogen production
GS4. express statements or information clearly so that others can understand
GS5. understand the main points of simple discussions
GS6. follow organization rule-based decision making process.
GS7. planning and organization of work to meet schedule.
GS8. work constructively and collaboratively with others.
GS9. communicate and create awareness.
GS10. recognize problems & approach relevant authority.
GS11. critically evaluate information obtained from supervisor and co-workers to perform day to day activities.
GS12. ask questions for better understanding.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Basics of Green Hydrogen Production	**30**	**20**	**-**	**-**
PC1. discuss properties and characteristics of Hydrogen	3	-	-	-
PC2. describe basic concepts of Hydrogen as fuel and energy carrier	3	-	-	-
PC3. discuss in brief various existing methods of hydrogen production and demonstrate various methods of hydrogen production.	4	2	-	-
PC4. discuss various colour code nomenclature of Hydrogen and demonstrate with chart colour code nomenclature of Hydrogen	4	3	-	-
PC5. discuss various technology options for production of Green Hydrogen and draw a flow diagram of green hydrogen production, conversion and end uses across the energy system	4	5	-	-
PC6. discuss key aspects and challenges related to production and storage of Green Hydrogen and do an activity for matching the process and source of production as applicable for different colour codes of hydrogen	4	5	-	-
PC7. discuss the end use applications of Green hydrogen in industry, transport and power production	4	-	-	-
PC8. discuss the role and responsibilities of various Green Hydrogen Plant Junior Technicians and demonstrate roles of various technicians in a green hydrogen plant	4	5	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4301
NOS Name	Basics of Green Hydrogen Production
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Hydrogen Plant Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4302: Analyse Main Parts of green hydrogen production unit

Description

This unit specifies identifying the main parts of green hydrogen production unit

Scope

The scope covers the following :

- Analyse Main Parts of green hydrogen production unit

Elements and Performance Criteria

Analyse Main Parts of green hydrogen production unit

To be competent, the user/individual on the job must be able to:

PC1. identify key parts and components of the Green Hydrogen plant including electrical, mechanical and civil components and illustrate the schematic of Green hydrogen production plant

PC2. discuss functions of each part and components and illustrate key components of the plant and outline their functions through plant schematic

PC3. discuss fundamental principles of main components on which they operate

PC4. explain basics of plant layout and illustrate how to interpret the Plant Layout including various equipments

PC5. discuss the plant components most relevant to the course and demonstrate how to interpret signs, notices and/or cautions at project site.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organizations reporting structure

KU2. organizations documentation policy

KU3. organizational culture

KU4. Properties and characteristic of hydrogen

KU5. concept of hydrogen as fuel and energy carrier

KU6. signs, symbols and color codes and nomenclature used in hydrogen production

KU7. challenges related to production and storage of green hydrogen

KU8. basic skills required to perform the task of green hydrogen production

KU9. understanding relevant regulations and safety standards

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. fill up relevant documents
GS2. read vernacular language
GS3. understand the various colour codes used in hydrogen production
GS4. express statements or information clearly so that others can understand
GS5. understand the main points of simple discussions
GS6. follow organization rule-based decision making process
GS7. planning and organization of work to meet schedule
GS8. work constructively and collaboratively with others
GS9. communicate and create awareness
GS10. ask questions for better understanding.
GS11. critically evaluate information obtained from supervisor and co-workers to perform day to day activities.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Analyse Main Parts of green hydrogen production unit	**30**	**20**	**-**	**-**
PC1. identify key parts and components of the Green Hydrogen plant including electrical, mechanical and civil components and illustrate the schematic of Green hydrogen production plant	6	5	-	-
PC2. discuss functions of each part and components and illustrate key components of the plant and outline their functions through plant schematic	6	5	-	-
PC3. discuss fundamental principles of main components on which they operate	6	5	-	-
PC4. explain basics of plant layout and illustrate how to interpret the Plant Layout including various equipments	6	-	-	-
PC5. discuss the plant components most relevant to the course and demonstrate how to interpret signs, notices and/or cautions at project site.	6	5	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4302
NOS Name	Analyse Main Parts of green hydrogen production unit
Sector	Green Jobs
Sub-Sector	Other Green Jobs
Occupation	Hydrogen Plant Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4311: Basics of Desalination process and Its Importance for Green Hydrogen Production

Description

This unit explain about basics of Desalination process and Its Importance for Green Hydrogen Production

Scope

The scope covers the following :

- Importance of desalination in green hydrogen production
- Desalination methodology and process involved

Elements and Performance Criteria

Basics of Desalination process and its importance in green hydrogen production

To be competent, the user/individual on the job must be able to:

PC1. discuss how desalination of water is important for green hydrogen production and illustrate desalination methodology along with their key features and their key specification through Pictures, videos, product data sheet etc

PC2. explain basics of desalination

PC3. describe an overview of desalination methodology and their key features and outline differences in various desalination technologies like thermal and membrane and illustrate their schematics

PC4. explain major components of a desalination unit

PC5. discuss the installation procedure of desalination unit and depict flowchart for desalination process in detail

PC6. discuss the installation procedure of pumps associated with desalination unit

PC7. discuss the Operating principles and procedure of desalination and demonstrate desalination process through pictures and videos

PC8. explain the concept of thermal and membrane technology

PC9. identify the key attributes of desalination process for successful production of green hydrogen in India and overseas and analyse the key attributes of desalination process for successful production of green hydrogen in India and overseas.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. company's installation policy

KU2. document information using appropriate corporate forms

KU3. obtain authorization from specified field safety officer and supervisor

KU4. relevant personal protective equipment required within the GH2 production facility

KU5. relevant standards and regulations to be followed in the GH2 production facility

KU6. occupational health and safety (OHS) standards to be followed in the GH2 production facility
KU7. risk identification and mitigation procedure for safe work in a GH2 generation facility
KU8. know how of tools & tackles required to carry out the work

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. fill up documentation applicable to one's role
GS2. read vernacular/english language
GS3. read and understand manuals, health and safety instructions, memos, other company documents
GS4. ability to read from different sources- books, screens in machines and signage
GS5. understand the various colour codes, as per standard electrical, mechanical and civil nomenclature
GS6. express statements or information clearly so that others can hear and understand
GS7. participate in and understand the main points of simple discussions
GS8. respond appropriately to any queries
GS9. communicate with peers, supervisor and sub-ordinates
GS10. follow organisation rule- based decision making process
GS11. take decision with systematic course of actions and/or response
GS12. plan and organize work schedule to meet deadlines
GS13. plan to utilise time and equipment's effectively
GS14. work constructively and collaboratively with others
GS15. follow organisational code of conduct
GS16. recognize problems and search for solutions
GS17. choose best methods to complete assigned tasks
GS18. approach relevant authority when required
GS19. apply domain knowledge, observations and data to select course of action to perform tasks related to the job role

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Basics of Desalination process and its importance in green hydrogen production	**30**	**20**	**-**	**-**
PC1. discuss how desalination of water is important for green hydrogen production and illustrate desalination methodology along with their key features and their key specification through Pictures, videos, product data sheet etc	3	3	-	-
PC2. explain basics of desalination	3	-	-	-
PC3. describe an overview of desalination methodology and their key features and outline differences in various desalination technologies like thermal and membrane and illustrate their schematics	3	3	-	-
PC4. explain major components of a desalination unit	3	-	-	-
PC5. discuss the installation procedure of desalination unit and depict flowchart for desalination process in detail	3	4	-	-
PC6. discuss the installation procedure of pumps associated with desalination unit	4	-	-	-
PC7. discuss the Operating principles and procedure of desalination and demonstrate desalination process through pictures and videos	3	5	-	-
PC8. explain the concept of thermal and membrane technology	5	-	-	-
PC9. identify the key attributes of desalination process for successful production of green hydrogen in India and overseas and analyse the key attributes of desalination process for successful production of green hydrogen in India and overseas.	3	5	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4311
NOS Name	Basics of Desalination process and Its Importance for Green Hydrogen Production
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4312: Perform pretreatment of water before Desalination for green hydrogen production

Description

This unit is about to perform pretreatment of water before Desalination for green hydrogen production

Scope

The scope covers the following :

- process of pre-treatment of water, Plant specifications
- step by step process of pre-treatment of water
- sub processes used in Pre-treatment of water

Elements and Performance Criteria

Describe the pretreatment process of water before desalination

To be competent, the user/individual on the job must be able to:

PC1. discuss about pre-treatment process of water before desalination and demonstrate the step-by-step procedure for pre-treatment of water on small model

PC2. discuss various techniques used for pre-treatment

PC3. discuss step by step process for pre-treating of water before sending it to desalination unit

PC4. explain the various processes like Coagulation, filtration, sedimentation, etc

PC5. discuss clarification process and demonstrate Clarification process through pictures and videos

PC6. discuss uses of activated carbon filter and Pressure Sand Filter

PC7. discuss why pre-treatment is necessary before desalination

PC8. discuss types of chemicals and reagents used to pre-treat water

PC9. discuss the operation of pumps used in pre-treating water and demonstrate lining up of various pumps and valves used for pre-treating water

PC10. identify successful case studies for pretreatment of water for green hydrogen production in India and overseas and analyse successful case studies for pretreatment of water for green hydrogen production in India and overseas

PC11. show how to read and interpret flowchart and process of desalination unit as per concerned technical sheets through pictures and videos

PC12. show how to perform requisite data and document management

PC13. show how to measure and monitor parameters of water at different stages of pre-treatment process

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. company's installation policy

KU2. document information using appropriate corporate forms
KU3. obtain authorization from specified field safety officer and supervisor
KU4. relevant personal protective equipments required within the facility
KU5. relevant standards and regulations to be followed in the facility
KU6. occupational health and safety (OHS) standards to be followed in the facility
KU7. risk identification and mitigation procedure for safe work in a facility
KU8. know how of tools & tackles required to carry out the work

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. fill up documentation applicable to one's role
GS2. read vernacular/english language
GS3. read and understand manuals, health and safety instructions, memos, other company documents
GS4. ability to read from different sources- books, screens in machines and signage
GS5. understand the various colour codes, as per standard electrical, mechanical and civil nomenclature
GS6. express statements or information clearly so that others can hear and understand
GS7. participate in and understand the main points of simple discussions
GS8. respond appropriately to any queries
GS9. communicate with peers, supervisor and sub-ordinates
GS10. follow organisation rule- based decision making process
GS11. take decision with systematic course of actions and/or response
GS12. plan and organize work schedule to meet deadlines
GS13. plan to utilise time and equipment's effectively
GS14. work constructively and collaboratively with others
GS15. follow organisational code of conduct
GS16. recognize problems and search for solutions
GS17. choose best methods to complete assigned tasks
GS18. apply domain knowledge, observations and data to select course of action to perform tasks related to the job role
GS19. critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Describe the pretreatment process of water before desalination	**29**	**21**	**-**	**-**
PC1. discuss about pre-treatment process of water before desalination and demonstrate the step-by-step procedure for pre-treatment of water on small model	2	3	-	-
PC2. discuss various techniques used for pre-treatment	3	-	-	-
PC3. discuss step by step process for pre-treating of water before sending it to desalination unit	3	-	-	-
PC4. explain the various processes like Coagulation, filtration, sedimentation, etc	3	-	-	-
PC5. discuss clarification process and demonstrate Clarification process through pictures and videos	3	3	-	-
PC6. discuss uses of activated carbon filter and Pressure Sand Filter	3	-	-	-
PC7. discuss why pre-treatment is necessary before desalination	3	-	-	-
PC8. discuss types of chemicals and reagents used to pre-treat water	3	-	-	-
PC9. discuss the operation of pumps used in pre-treating water and demonstrate lining up of various pumps and valves used for pre-treating water	3	3	-	-
PC10. identify successful case studies for pretreatment of water for green hydrogen production in India and overseas and analyse successful case studies for pretreatment of water for green hydrogen production in India and overseas	3	3	-	-
PC11. show how to read and interpret flowchart and process of desalination unit as per concerned technical sheets through pictures and videos	-	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. show how to perform requisite data and document management	-	3	-	-
PC13. show how to measure and monitor parameters of water at different stages of pre-treatment process	-	3	-	-
NOS Total	**29**	**21**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4312
NOS Name	Perform pretreatment of water before Desalination for green hydrogen production
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4313: Quality of water feed to the electrolyser

Description

This unit describe about quality of water feed to the electrolyser.

Scope

The scope covers the following :

- Discuss about the quality of input water feed to the electrolyser
- Important parameters used to measure water quality
- Describe Tools and Tackles used in desalination process

Elements and Performance Criteria

Discuss about the quality of input water feed to the electrolyser

To be competent, the user/individual on the job must be able to:

PC1. explain parameters of water feeding to electrolyser conforming to relevant technical sheets, safety and technical standards for proper execution of work and demonstrate suitable parameters of water feeding to electrolyser conforming to relevant technical sheets, safety and technical standards.

PC2. discuss about Reverse Osmosis Process and demonstrate Reverse Osmosis Process

PC3. discuss about water Pressure required for Reverse Osmosis

PC4. discuss various parameters like temperature, dissolved oxygen, pH, conductivity, ORP, and turbidity that are frequently sampled or monitored

PC5. discuss the Regeneration Process of water

PC6. discuss the time required for Regeneration

PC7. discuss the chemicals used in Regeneration Process and show the type of chemicals and reagents used to maintain quality

PC8. discuss about Brine Blowdown in desalination

PC9. discuss about treatment and disposal of brine

PC10. discuss the process of maintaining feed water quality and demonstrate the consequences of poor water quality

PC11. discuss the type of chemical and reagents used to maintain quality

PC12. discuss about the MSDS of chemicals used for maintaining water quality

PC13. discuss the importance for maintaining quality of water before feeding to electrolyser

PC14. identify key attributes of water quality feed to the electrolyser for green hydrogen production in India and overseas and analyse key attributes of water quality feed to the electrolyser for green hydrogen production in India and overseas

PC15. show the working of various devices used to measure various parameters like PH, CONDUCTIVITY, TDS of water before feeding to electrolyser

PC16. demonstrate the consequences like scaling and fouling through pictures and videos

Describe Tools and Tackles used in desalination process

To be competent, the user/individual on the job must be able to:

PC17. explain the tools and tackles required in desalination unit at green hydrogen generation project site and demonstrate the tools and tackles required in desalination unit at green hydrogen generation project site.

PC18. explain the working of tools and tackles used at desalination unit and demonstrate the safe working of different types of tools and tackles used at desalination unit

PC19. describe how to avoid use of damaged tools while working at desalination unit and show how to avoid potential causes of using damaged tools while working at desalination unit.

PC20. discuss the devices used for measuring water parameters at desalination unit and demonstrate how to measure pressure, temperature and flow of water

PC21. discuss checking and detection of damaged tools and show how to check and detect damaged tools

PC22. discuss working procedure of each tool required for handling desalination unit

PC23. discuss the routine inspection carried out to check the tools are corrosion free

PC24. discuss about the device used for measuring PH of water and demonstrate how to measure PH.

PC25. discuss about the device used for measuring conductivity of water demonstrate how to measure Conductivity

PC26. discuss about the device used for measuring silica in water and demonstrate how to measure Silica

PC27. discuss types of flowmeters, valves, control valves used in desalination unit and demonstrate different types of flowmeters, valves, control valves used in desalination unit through pictures or videos

PC28. identify best quality of tools and tackles used in desalination of water for green hydrogen production in India and overseas and showcase best quality of tools and tackles used in desalination of water for green hydrogen production in India and overseas through picture or videos.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. company's installation policy

KU2. document information using appropriate corporate forms

KU3. obtain authorization from specified field safety officer and supervisor

KU4. relevant personal protective equipments required within the manufacturing facility

KU5. relevant standards and regulations to be followed in the facility

KU6. occupational health and safety (OHS) standards to be followed in the facility

KU7. risk identification and mitigation procedure for safe work in a facility

KU8. know how of tools & tackles required to carry out the work

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. fill up documentation applicable to one's role

GS2. read vernacular/english language

GS3. read and understand manuals, health and safety instructions, memos, other company documents

GS4. read and understand manuals, health and safety instructions, memos, other company documents

GS5. ability to read from different sources- books, screens in machines and signage

GS6. understand the various colour codes, as per standard electrical, mechanical and civil nomenclature

GS7. express statements or information clearly so that others can hear and understand

GS8. participate in and understand the main points of simple discussions

GS9. respond appropriately to any queries

GS10. communicate with peers, supervisor and sub-ordinates

GS11. follow organisation rule- based decision making process

GS12. take decision with systematic course of actions and/or response

GS13. plan and organize work schedule to meet deadlines

GS14. plan to utilise time and equipment's effectively

GS15. work constructively and collaboratively with others

GS16. recognize problems and search for solutions

GS17. choose best methods to complete assigned tasks

GS18. apply domain knowledge, observations and data to select course of action to perform tasks related to the job role

GS19. critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Discuss about the quality of input water feed to the electrolyser	**15**	**13**	**-**	**-**
PC1. explain parameters of water feeding to electrolyser conforming to relevant technical sheets, safety and technical standards for proper execution of work and demonstrate suitable parameters of water feeding to electrolyser conforming to relevant technical sheets, safety and technical standards.	2	2	-	-
PC2. discuss about Reverse Osmosis Process and demonstrate Reverse Osmosis Process	1	1	-	-
PC3. discuss about water Pressure required for Reverse Osmosis	1	-	-	-
PC4. discuss various parameters like temperature, dissolved oxygen, pH, conductivity, ORP, and turbidity that are frequently sampled or monitored	1	-	-	-
PC5. discuss the Regeneration Process of water	1	-	-	-
PC6. discuss the time required for Regeneration	1	-	-	-
PC7. discuss the chemicals used in Regeneration Process and show the type of chemicals and reagents used to maintain quality	1	2	-	-
PC8. discuss about Brine Blowdown in desalination	1	-	-	-
PC9. discuss about treatment and disposal of brine	1	-	-	-
PC10. discuss the process of maintaining feed water quality and demonstrate the consequences of poor water quality	1	2	-	-
PC11. discuss the type of chemical and reagents used to maintain quality	1	-	-	-
PC12. discuss about the MSDS of chemicals used for maintaining water quality	1	-	-	-
PC13. discuss the importance for maintaining quality of water before feeding to electrolyser	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. identify key attributes of water quality feed to the electrolyser for green hydrogen production in India and overseas and analyse key attributes of water quality feed to the electrolyser for green hydrogen production in India and overseas	1	2	-	-
PC15. show the working of various devices used to measure various parameters like PH, CONDUCTIVITY, TDS of water before feeding to electrolyser	-	2	-	-
PC16. demonstrate the consequences like scaling and fouling through pictures and videos	-	2	-	-
Describe Tools and Tackles used in desalination process	**12**	**10**	**-**	**-**
PC17. explain the tools and tackles required in desalination unit at green hydrogen generation project site and demonstrate the tools and tackles required in desalination unit at green hydrogen generation project site.	1	1	-	-
PC18. explain the working of tools and tackles used at desalination unit and demonstrate the safe working of different types of tools and tackles used at desalination unit	1	1	-	-
PC19. describe how to avoid use of damaged tools while working at desalination unit and show how to avoid potential causes of using damaged tools while working at desalination unit.	1	1	-	-
PC20. discuss the devices used for measuring water parameters at desalination unit and demonstrate how to measure pressure, temperature and flow of water	1	1	-	-
PC21. discuss checking and detection of damaged tools and show how to check and detect damaged tools	1	1	-	-
PC22. discuss working procedure of each tool required for handling desalination unit	1	-	-	-
PC23. discuss the routine inspection carried out to check the tools are corrosion free	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. discuss about the device used for measuring PH of water and demonstrate how to measure PH.	1	1	-	-
PC25. discuss about the device used for measuring conductivity of water demonstrate how to measure Conductivity	1	1	-	-
PC26. discuss about the device used for measuring silica in water and demonstrate how to measure Silica	1	1	-	-
PC27. discuss types of flowmeters, valves, control valves used in desalination unit and demonstrate different types of flowmeters, valves, control valves used in desalination unit through pictures or videos	1	1	-	-
PC28. identify best quality of tools and tackles used in desalination of water for green hydrogen production in India and overseas and showcase best quality of tools and tackles used in desalination of water for green hydrogen production in India and overseas through picture or videos.	1	1	-	-
NOS Total	**27**	**23**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4313
NOS Name	Quality of water feed to the electrolyser
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4314: Perform Operation & Maintenance of desalination unit

Description

This unit Perform Operation & Maintenance of desalination unit

Scope

The scope covers the following :

- Important key parameters related to operations of Desalination Plant
- Preventive maintenance related activities in details

Elements and Performance Criteria

Operation & Maintenance of desalination unit

To be competent, the user/individual on the job must be able to:

PC1. discuss operations related to smooth running of desalination unit and show how to operate the desalination unit as per concerned technical sheets through pictures or videos

PC2. discuss standard operating procedure of Desalination unit and demonstrate the operation of desalination unit through pictures or videos

PC3. discuss various operating parameters required to monitor desalination unit and demonstrate measurement of various parameters like PH, Conductivity, Silica, Turbidity in desalination unit

PC4. explain the operation of various pumps and valves and demonstrate the operating procedure of valves, control valves, pumps etc

PC5. discuss types of chemicals and reagents used in water during desalination and show how to use different types of chemical and reagents for treating water in desalination unit through pictures and videos

PC6. discuss the measurement units related to water parameters in desalination unit and show how to read and monitor important parameters of water in desalination unit

PC7. discuss how to do Preventive maintenance in desalination unit

PC8. discuss the process of replacing resins and membranes in desalination unit

PC9. discuss the operation of various devices used to measure water parameters

PC10. discuss process for storage and distribution of desalinated water to electrolyser

PC11. discuss operation of various types of pumps and valves used in pre-treatment of water

PC12. discuss the operation of blowdown of brine

PC13. explain the concept of zero liquid leakage in Desalination unit.

PC14. identify best practices for safe and successful operation and maintenance of Desalination unit for green hydrogen production in India and overseas and analyse best practices for safe and successful operation and maintenance of Desalination unit for green hydrogen production in India and overseas.

PC15. show how to perform requisite data and document management

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. company's installation policy
KU2. company's customer support policy
KU3. document information using appropriate corporate forms
KU4. obtain authorization from specified field safety officer and supervisor
KU5. relevant personal protective equipments required within the manufacturing facility
KU6. relevant standards and regulations to be followed in the manufacturing facility
KU7. occupational health and safety (OHS) standards to be followed in the manufacturing facility
KU8. risk identification and mitigation procedure for safe work in a facility
KU9. know how of tools & tackles required to carry out the work

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. fill up documentation applicable to one's role
GS2. read vernacular/english language
GS3. read and understand manuals, health and safety instructions, memos, other company documents
GS4. ability to read from different sources- books, screens in machines and signage
GS5. understand the various colour codes, as per standard electrical, mechanical and civil nomenclature
GS6. express statements or information clearly so that others can hear and understand
GS7. participate in and understand the main points of simple discussions
GS8. respond appropriately to any queries
GS9. communicate with peers, supervisor and sub-ordinates
GS10. follow organisation rule- based decision making process
GS11. take decision with systematic course of actions and/or response
GS12. plan and organize work schedule to meet deadlines
GS13. plan to utilise time and equipment's effectively
GS14. work constructively and collaboratively with others
GS15. follow organisational code of conduct
GS16. recognize problems and search for solutions
GS17. choose best methods to complete assigned tasks
GS18. apply domain knowledge, observations and data to select course of action to perform tasks related to the job role
GS19. critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Operation & Maintenance of desalination unit	**30**	**20**	**-**	**-**
PC1. discuss operations related to smooth running of desalination unit and show how to operate the desalination unit as per concerned technical sheets through pictures or videos	2	2	-	-
PC2. discuss standard operating procedure of Desalination unit and demonstrate the operation of desalination unit through pictures or videos	2	2	-	-
PC3. discuss various operating parameters required to monitor desalination unit and demonstrate measurement of various parameters like PH, Conductivity, Silica, Turbidity in desalination unit	2	2	-	-
PC4. explain the operation of various pumps and valves and demonstrate the operating procedure of valves, control valves, pumps etc	2	2	-	-
PC5. discuss types of chemicals and reagents used in water during desalination and show how to use different types of chemical and reagents for treating water in desalination unit through pictures and videos	2	2	-	-
PC6. discuss the measurement units related to water parameters in desalination unit and show how to read and monitor important parameters of water in desalination unit	2	2	-	-
PC7. discuss how to do Preventive maintenance in desalination unit	2	-	-	-
PC8. discuss the process of replacing resins and membranes in desalination unit	2	-	-	-
PC9. discuss the operation of various devices used to measure water parameters	2	-	-	-
PC10. discuss process for storage and distribution of desalinated water to electrolyser	2	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. discuss operation of various types of pumps and valves used in pre-treatment of water	3	-	-	-
PC12. discuss the operation of blowdown of brine	3	-	-	-
PC13. explain the concept of zero liquid leakage in Desalination unit.	3	-	-	-
PC14. identify best practices for safe and successful operation and maintenance of Desalination unit for green hydrogen production in India and overseas and analyse best practices for safe and successful operation and maintenance of Desalination unit for green hydrogen production in India and overseas.	1	4	-	-
PC15. show how to perform requisite data and document management	-	4	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4314
NOS Name	Perform Operation & Maintenance of desalination unit
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4315: Maintain Health & Safety at Desalination unit of Green Hydrogen generation project site

Description

This unit is about to Maintain Health & Safety at Desalination unit of Green Hydrogen generation project site

Scope

The scope covers the following :

- Requirements for safe work area while handling desalination unit at Green hydrogen generation project site.
- Hazards associated while working on desalination unit with in a hydrogen generation system and their mitigation measures

Elements and Performance Criteria

Maintain Personal Health & Safety while handling the Desalination system.

To be competent, the user/individual on the job must be able to:

PC1. explain the requirements for safe work area at desalination unit in green hydrogen generation project site and demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at desalination unit in green hydrogen generation site

PC2. identify the personal protective equipment used for the specific purpose and demonstrate the usage of personal protective equipment for ensuring safety during installation and O&M work of desalination unit.

PC3. explain the importance of Occupational health & Safety standards and regulations for Basic considerations for the safety of Desalination systems

PC4. describe potential causes of emergency such as, acid leak, fire, explosion, bomb threatening, natural calamities etc and demonstrate the use of fire extinguishers, fire detection and alarm system.

PC5. discuss importance of different sensors, detectors and safety tools

PC6. discuss and review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures

PC7. explain the importance of administering first aid and demonstrate how to administer first aid

PC8. identify the hazards associated with desalination unit and hydrogen generation system

PC9. identify work safety procedures and instructions for working at Desalination plant complying with applicable safety regulations and show how to comply with all applicable statutory requirements along with safety regulations in terms of fire protection

PC10. discuss Mock testing of firefighting system

PC11. discuss all applicable statutory requirements along with safety regulations in terms of fire protection

PC12. explain how to incorporate good housekeeping practices and infection control guidelines and demonstrate good housekeeping and infection control & prevention practices

PC13. identify best practices for maintaining health and safety in Desalination unit for green hydrogen production in India and overseas and showcase best practices for maintaining health and safety in Desalination unit for green hydrogen production in India and overseas.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organizations reporting structure
KU2. organizations documentation policy
KU3. organizational culture
KU4. understanding relevant regulations and safety standards
KU5. signs, symbols and color codes and nomenclature used in hydrogen production
KU6. Properties and characteristic of hydrogen
KU7. concept of hydrogen as fuel and energy carrier

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. company's installation policy
GS2. document information using appropriate corporate forms
GS3. obtain authorization from specified field safety officer and supervisor
GS4. relevant personal protective equipments required within the facility
GS5. relevant standards and regulations to be followed in the facility
GS6. occupational health and safety (OHS) standards to be followed in the facility
GS7. risk identification and mitigation procedure for safe work in a facility
GS8. follow organization rule-based decision making process
GS9. planning and organization of work to meet schedule
GS10. work constructively and collaboratively with others.
GS11. communicate and create awareness

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Maintain Personal Health & Safety while handling the Desalination system.	**30**	**20**	**-**	**-**
PC1. explain the requirements for safe work area at desalination unit in green hydrogen generation project site and demonstrate how to follow necessary and adequate safety measures including personal protective equipment and precautions to avoid any accident at desalination unit in green hydrogen generation site	2	3	-	-
PC2. identify the personal protective equipment used for the specific purpose and demonstrate the usage of personal protective equipment for ensuring safety during installation and O&M work of desalination unit.	2	3	-	-
PC3. explain the importance of Occupational health & Safety standards and regulations for Basic considerations for the safety of Desalination systems	2	-	-	-
PC4. describe potential causes of emergency such as, acid leak, fire, explosion, bomb threatening, natural calamities etc and demonstrate the use of fire extinguishers, fire detection and alarm system.	2	3	-	-
PC5. discuss importance of different sensors, detectors and safety tools	2	-	-	-
PC6. discuss and review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures	3	-	-	-
PC7. explain the importance of administering first aid and demonstrate how to administer first aid	2	3	-	-
PC8. identify the hazards associated with desalination unit and hydrogen generation system	3	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. identify work safety procedures and instructions for working at Desalination plant complying with applicable safety regulations and show how to comply with all applicable statutory requirements along with safety regulations in terms of fire protection	2	3	-	-
PC10. discuss Mock testing of firefighting system	3	-	-	-
PC11. discuss all applicable statutory requirements along with safety regulations in terms of fire protection	2	-	-	-
PC12. explain how to incorporate good housekeeping practices and infection control guidelines and demonstrate good housekeeping and infection control & prevention practices	2	2	-	-
PC13. identify best practices for maintaining health and safety in Desalination unit for green hydrogen production in India and overseas and showcase best practices for maintaining health and safety in Desalination unit for green hydrogen production in India and overseas.	3	3	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4315
NOS Name	Maintain Health & Safety at Desalination unit of Green Hydrogen generation project site
Sector	Green Jobs
Sub-Sector	Green Hydrogen
Occupation	Hydrogen Plant Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

KU10. how to compute income and expenses
KU11. importance of maintaining safety and security in financial transactions
KU12. different legal rights and laws
KU13. how to operate digital devices and applications safely and securely
KU14. ways to identify business opportunities
KU15. types of customers and their needs
KU16. how to apply for a job and prepare for an interview
KU17. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively using appropriate language
GS2. behave politely and appropriately with all
GS3. perform basic calculations
GS4. solve problems effectively
GS5. be careful and attentive at work
GS6. use time effectively
GS7. maintain hygiene and sanitisation to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**1**	**3**	**-**	**-**
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
Communication Skills	**1**	**1**	**-**	**-**
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**1**	**-**	**-**
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
Financial and Legal Literacy	**3**	**4**	**-**	**-**
PC9. use various financial products and services safely and securely	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
Essential Digital Skills	**4**	**6**	**-**	**-**
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
Entrepreneurship	**3**	**5**	**-**	**-**
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
Customer Service	**2**	**2**	**-**	**-**
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**1**	**3**	**-**	**-**
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the Qualification assessment, every trainee should score the Recommended Pass % aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4301.Basics of Green Hydrogen Production	30	20	-	-	50	12
SGJ/N4302.Analyse Main Parts of green hydrogen production unit	30	20	-	-	50	12
SGJ/N4311.Basics of Desalination process and Its Importance for Green Hydrogen Production	30	20	0	0	50	12
SGJ/N4312.Perform pretreatment of water before Desalination for green hydrogen production	29	21	0	0	50	12
SGJ/N4313.Quality of water feed to the electrolyser	27	23	0	0	50	12
SGJ/N4314.Perform Operation & Maintenance of desalination unit	30	20	0	0	50	12
SGJ/N4315.Maintain Health & Safety at Desalination unit of Green Hydrogen generation project site	30	20	0	0	50	12
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	16
Total	**226**	**174**	**-**	**-**	**400**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.